

The left side of the slide features several overlapping, semi-transparent blue geometric shapes. These include a large triangle pointing downwards, a smaller triangle pointing upwards, and a rounded rectangle. The shapes are arranged in a way that creates a sense of depth and movement.

# **Intro to AI Agents**

## **Week 3**

# Last Week

Week 1: Introduction to Generative AI

From AI to Generative AI. Use cases.

Week 2: RAG

Lilli Case Study, RAG

Lilli



GenAI Guidelines

Contact Us

What's New

Knowledge Base



Client Capabilities

GenAI

## Welcome to Lilli

Create a company profile

Explain a concept

Draft a template email

Structure a text to MICE categories

View more prompts

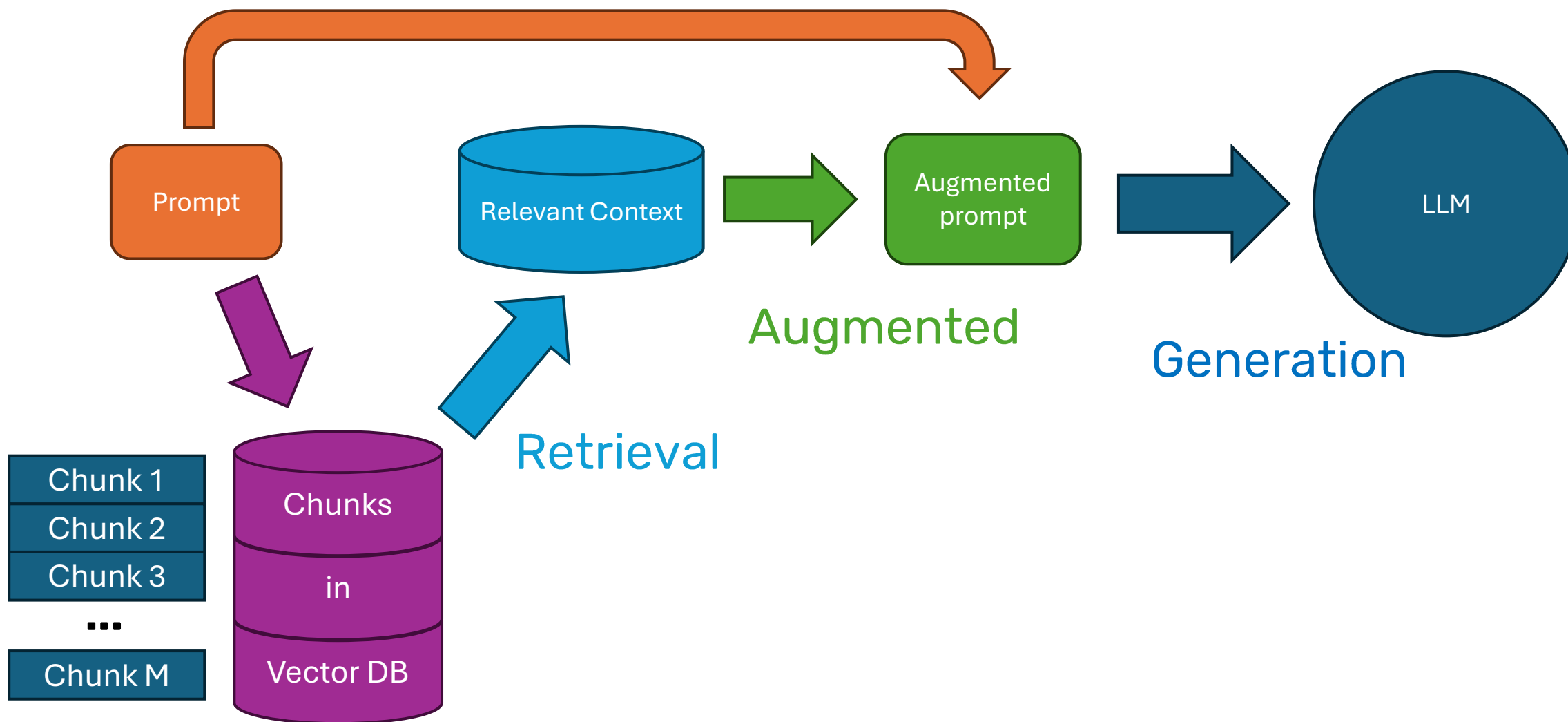
Not sure where to start? [How to write a great prompt](#) →

Enter your prompt



Lilli may produce inaccurate information about people, places, or facts. Please refer to the Lilli's guidelines on use of Generative AI.

# Retrieval Augmented Generation



# RAG: Pros and Cons

## Pros

Up to date information

Reduce hallucinations

Cheaper then fine-tuning

Domain specific

Data Privacy

Scalable

Explainability

## Cons

Increase latency

Complex

Cost

**Retrieval Quality?**

# This Week

## Week 1: Introduction to Generative AI

From AI to Generative AI. Use cases.

## Week 2: RAG

Lilli Case Study, RAG

## Week 3: Intro to AI Agents

Cont.' RAG, AI Agents, Reasoning

# RAG Quality

# RAG: Pros and Cons

## Pros

Up to date information

Reduce hallucinations

Cheaper then fine-tuning

Domain specific

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Explainability

## Cons

Increase latency

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**Retrieval Quality?**

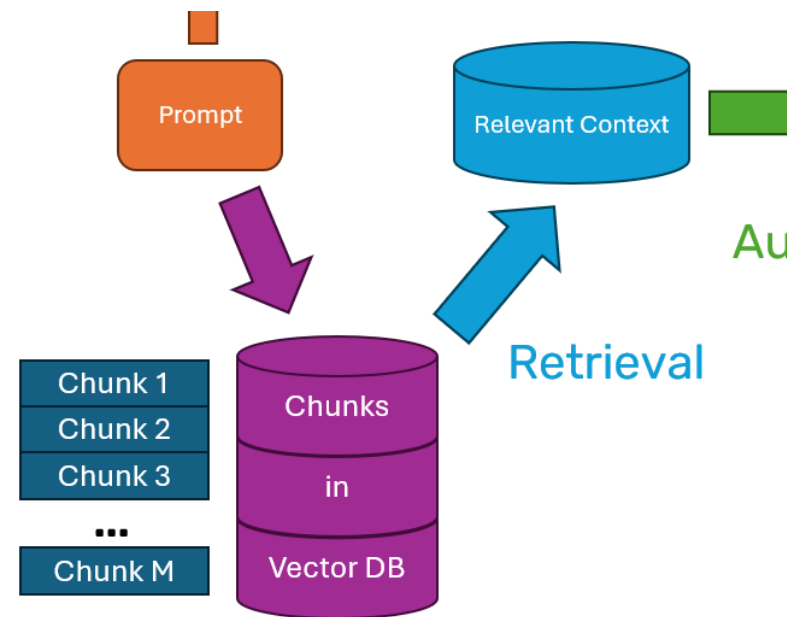


How to **retrieve** the most relevant context?

**How to effectively embed documents?**

**How much context to retrieve?**

**What Content Should Be Included in a RAG Dataset?**



# How to effectively embed documents?

Should I embed full documents?

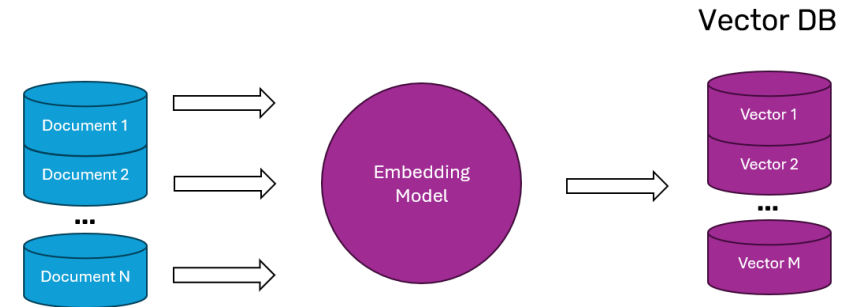
Risk of meaning becoming “blurry”  
for long documents

Should I embed each word/sentence?

Risk of losing context

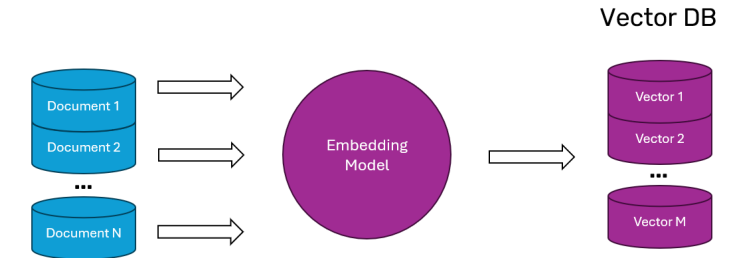
Should each chunk be independent?

Risk of losing coherency between chunks



# Best practices in embedding documents

## Chunk size:



| Data type                     | Tokens per chunk | Why?                                 |
|-------------------------------|------------------|--------------------------------------|
| General, long articles, books | 512 – 1024+      | Standard for most use cases          |
| Code & technical Docs         | 100 – 512        | Keeps concepts tight + reduces noise |
| Conversational data (chat)    | 200-400          | Natural conversational segments      |

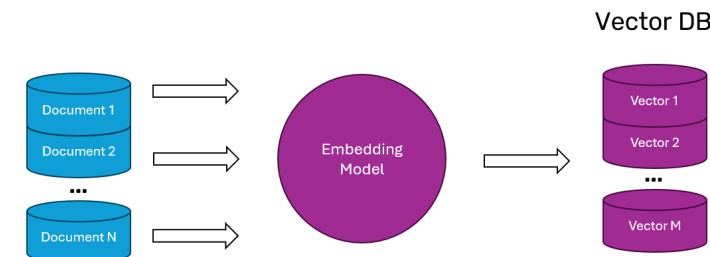
## General Rules

Too small => more focused but loses context

Too big => richer context but less focused

# Trade offs in embedding documents

## Overlap:



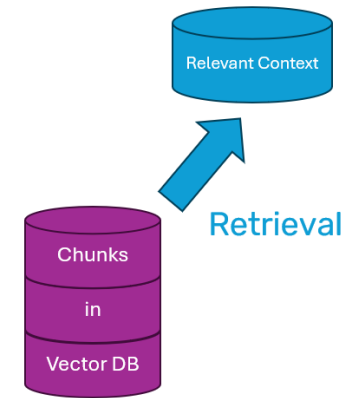
| Data type                   | Percentage of chunk size | Why?                                  |
|-----------------------------|--------------------------|---------------------------------------|
| General text                | 10-20%                   | Keeps paragraphs connected            |
| Long articles, books        | 5-15%                    | Less expensive                        |
| Code & technical Docs       | 30-50%                   | High dependability between code lines |
| Structured docs (i.e legal) | 15-25%                   | Prevents cutting clauses in half      |

## General Rules

Too small => breaks context between chunks

Too big => unnecessary duplication + higher cost

# How much context to retrieve?



## Top K:

| Data type                  | Chunks to retrieve | Why?                                     |
|----------------------------|--------------------|------------------------------------------|
| General Text               | 3-5                | Balanced relevance and cost              |
| Research Papers, long text | 8-12               | Answers require broad context            |
| Code & Technical Docs      | 10-20              | Code logic can be scattered across files |

## General Rules

Too small => missing key context, incomplete answers

Too big => additional noise, can cause hallucinations

**Type of Data:**    **Long Forms:** User Manuals, API Docs, etc.

**Good for RAG**

**Bad for RAG**

**Type of Data:    Research papers and academic content**

**Good for RAG**

**Long Forms:** User Manuals, API Docs, etc.

**Bad for RAG**

**Type of Data:**    **Real time data:** Stock prices, traffic updates

## Good for RAG

**Long Forms:** User Manuals, API Docs, etc.

**Research papers and academic content**

## Bad for RAG



## Type of Data:    **Code Repositories & Documentation**

### **Good for RAG**

**Long Forms:** User Manuals, API Docs, etc.

**Research papers and academic content**

### **Bad for RAG**

**Real time data:** Stock prices, traffic updates

**Type of Data:    Inventory databases with quantities**

**Good for RAG**

**Long Forms:** User Manuals, API Docs, etc.

**Research papers and academic content**

**Code repositories & Documentation**

**Bad for RAG**

**Real time data:** Stock prices, traffic updates

**Type of Data: Businesses locations for traffic app**

**Good for RAG**

**Long Forms:** User Manuals, API Docs, etc.

**Research papers and academic content**

**Code repositories & Documentation**

**Bad for RAG**

**Real time data:** Stock prices, traffic updates

**Inventory databases with quantities**

## Type of Data:

### Good for RAG

**Long Forms:** User Manuals, API Docs, etc.

**Research papers and academic content**

**Code repositories & Documentation**

### Bad for RAG

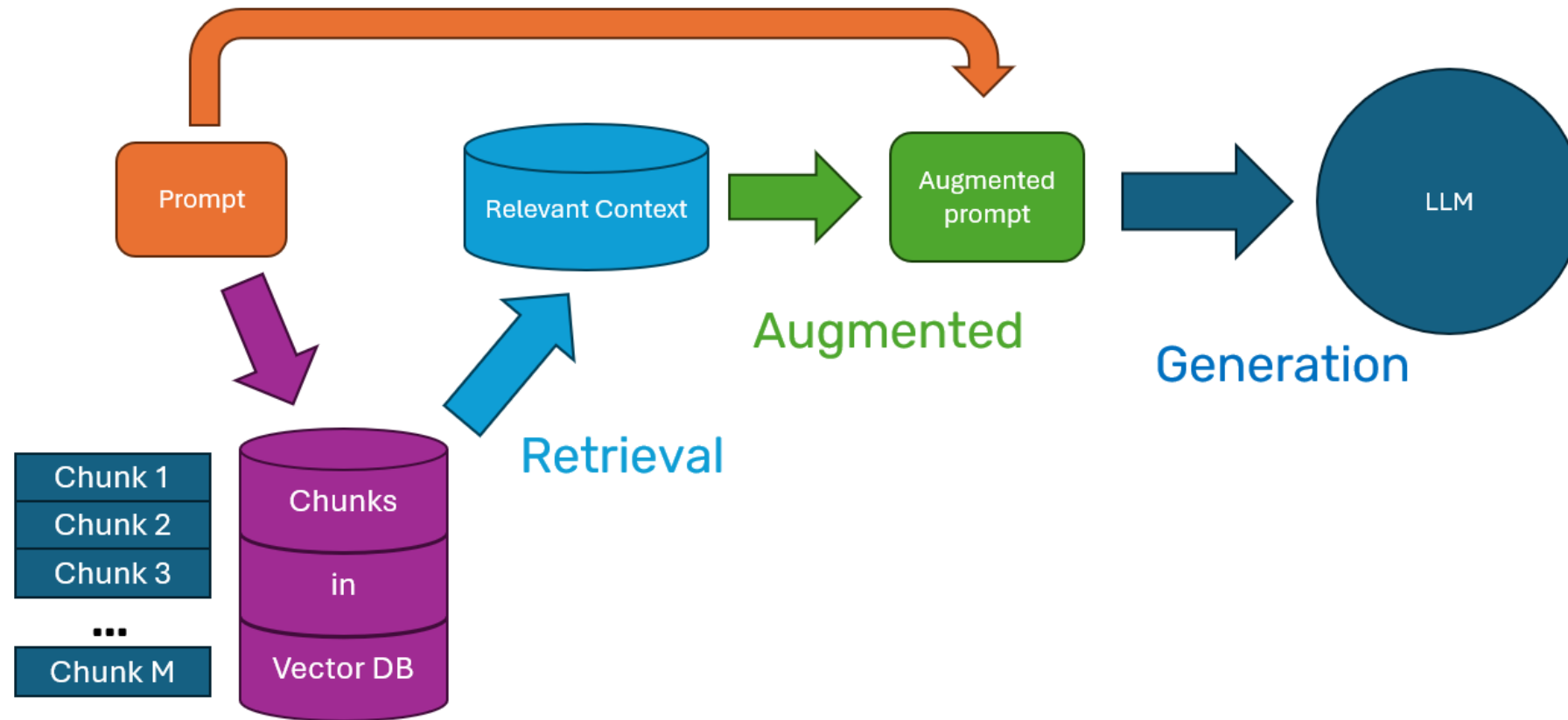
**Real time data:** Stock prices, traffic updates

**Inventory databases with quantities**

**Businesses locations for traffic app**

# Intro to AI Agents

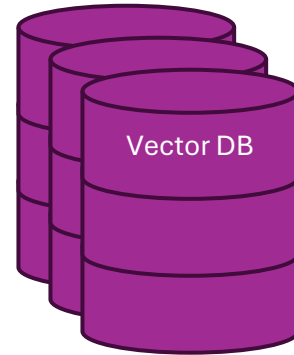
# Typical RAG Flow



\* Using the LLM only once to generate response

# What if

Using several Vector DBs?



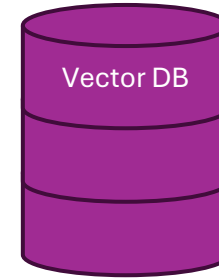
Several type of responses: Text, doc, code, image, video?



# Example (Consulting Tech)

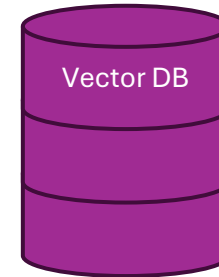
Client A (Yamato Heavy Industries):

A traditional car manufacturer trying to transition to EVs.



Client B (EcoAuto):

A futuristic transportation startup focusing on autonomous vehicles.





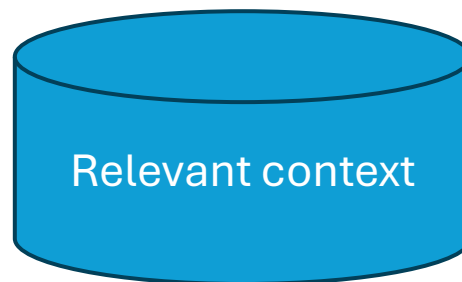
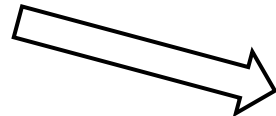
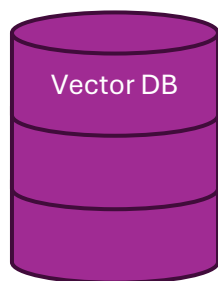
Summarize the **Q3 assembly line output** specifically from **Yamato Heavy Industries'** Kyoto plant logs.

+  Tools

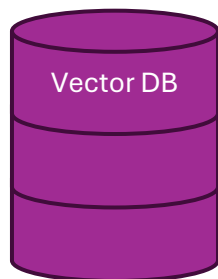
Thinking ▾



## Yamato Heavy Industries



## EcoAuto



Calculate the **monthly cash burn rate** based on the financial tables in EcoAuto's latest Series-A investor deck.

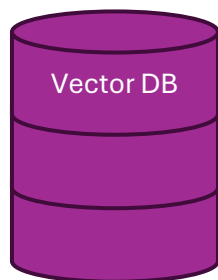


Tools

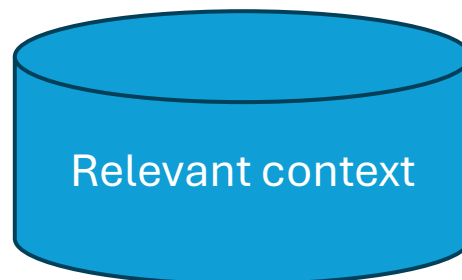
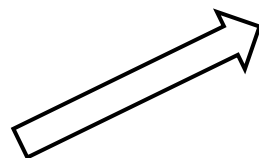
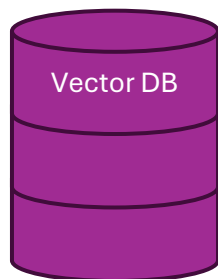
Thinking ▾



## Yamato Heavy Industries



EcoAuto



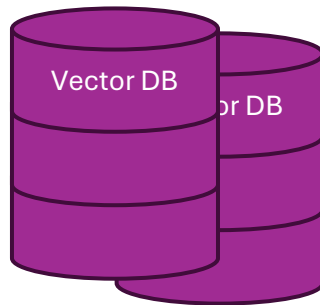
Explain the definition of 'Operational Leverage' as described in our firm's **internal 'Global Strategy Playbook**

+  Tools

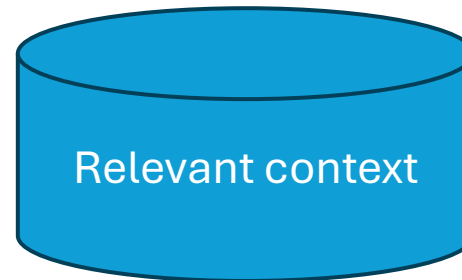
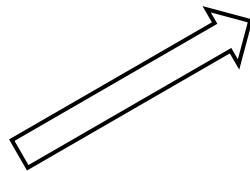
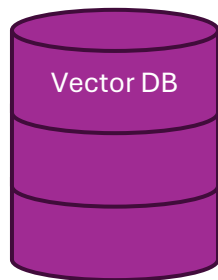
Thinking ▾



## Clients



## General Knowledge



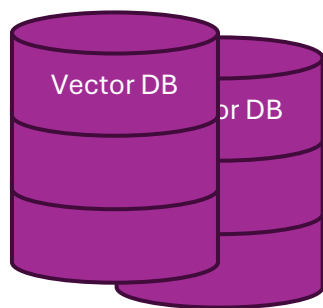
Retrieve **Toyota's confidential internal engineering logs** to see how they solved the specific engine failure Yamato is currently facing.

+ 🛠️ Tools

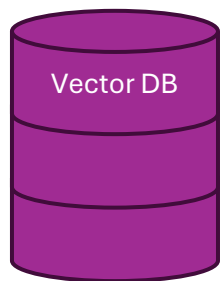
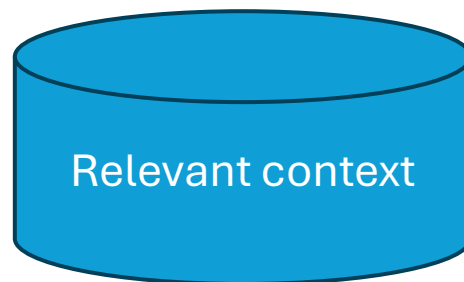
Thinking ▾



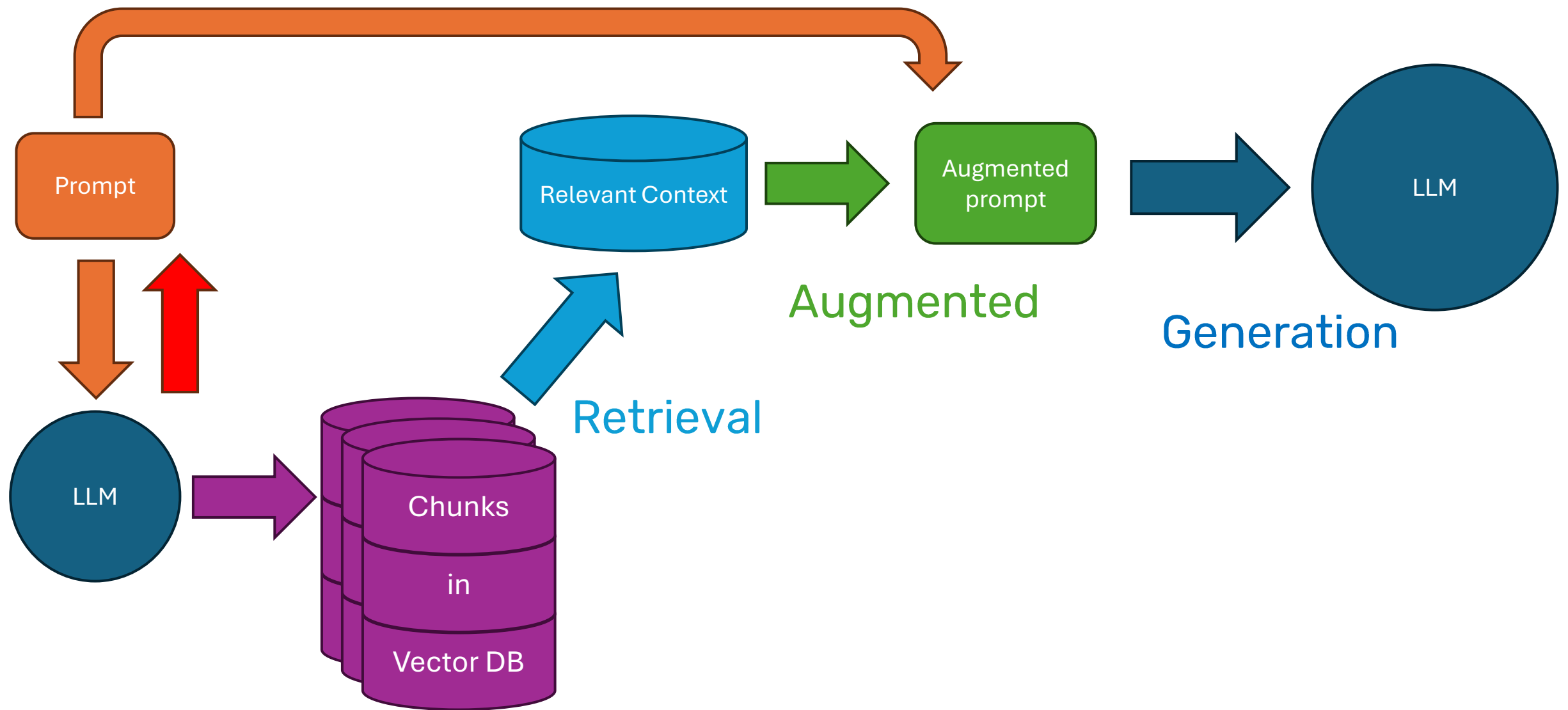
## Clients



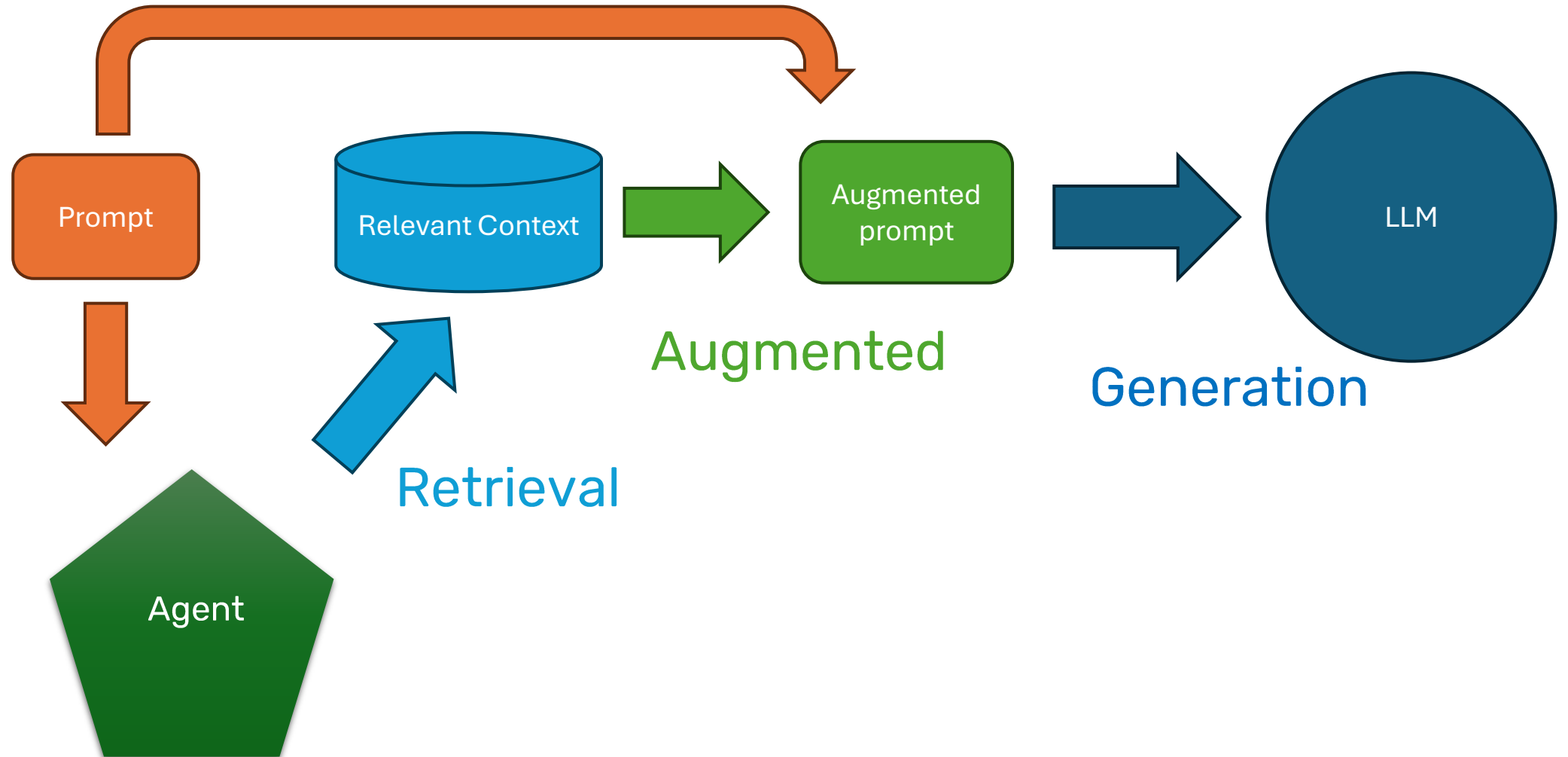
General Knowledge



# A solution

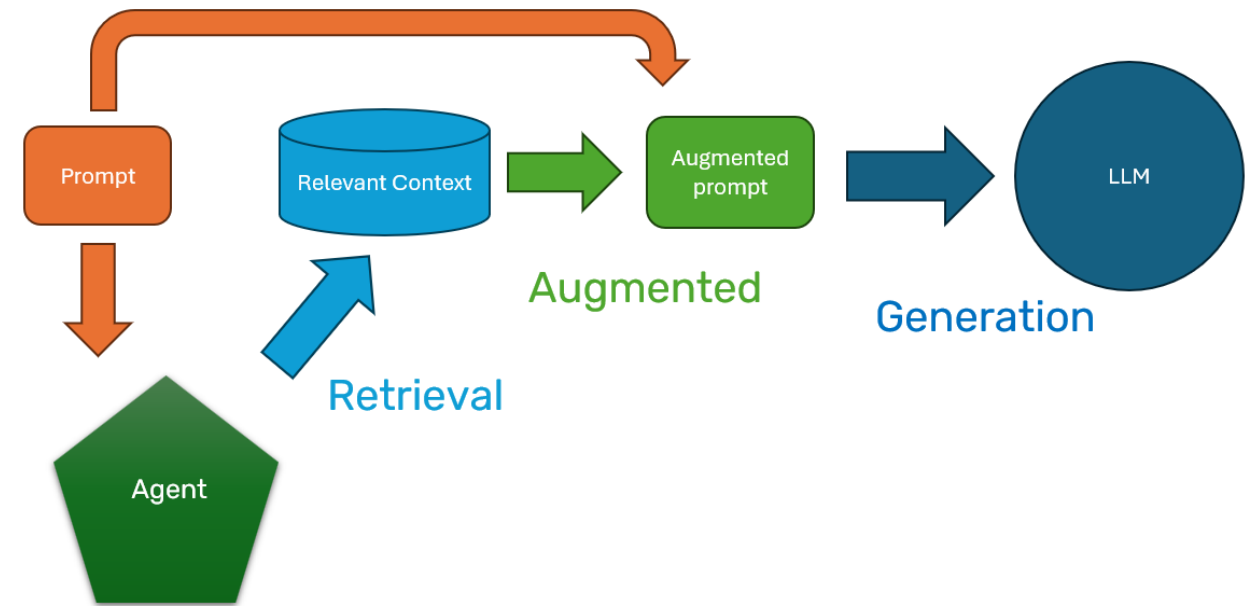


# A RAG Variant – Agentic RAG



# Agentic RAG

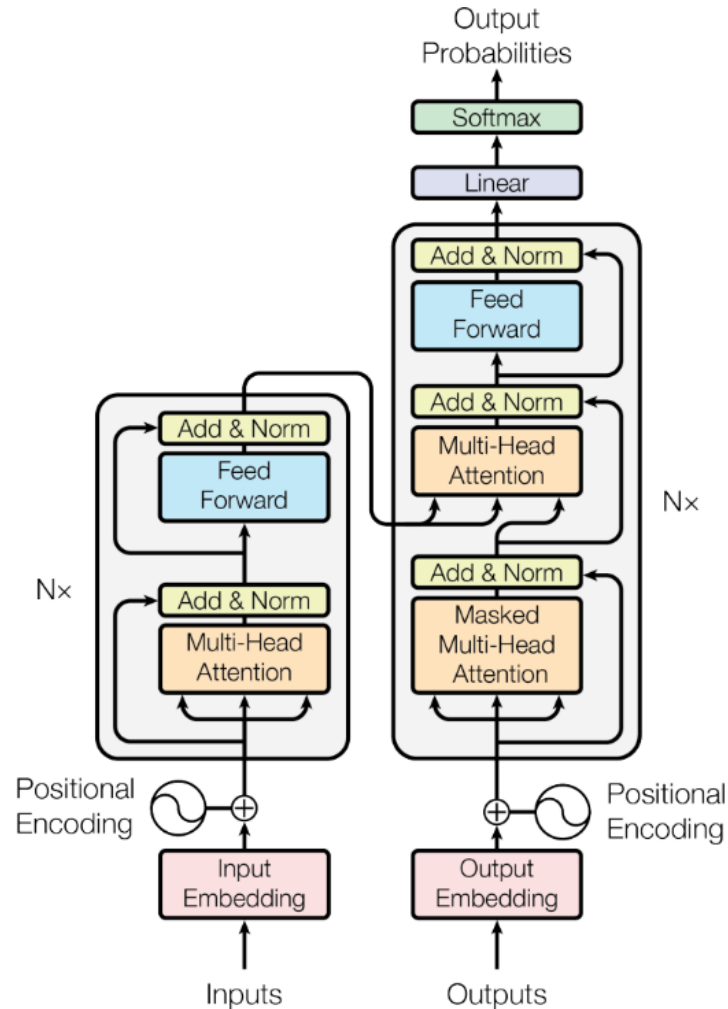
A smarter version of RAG that **reasons** through the task, refine the prompt, retrieve documents, and build a better final response



How can the LLM do that?



# 2017 Invention of the Transformer



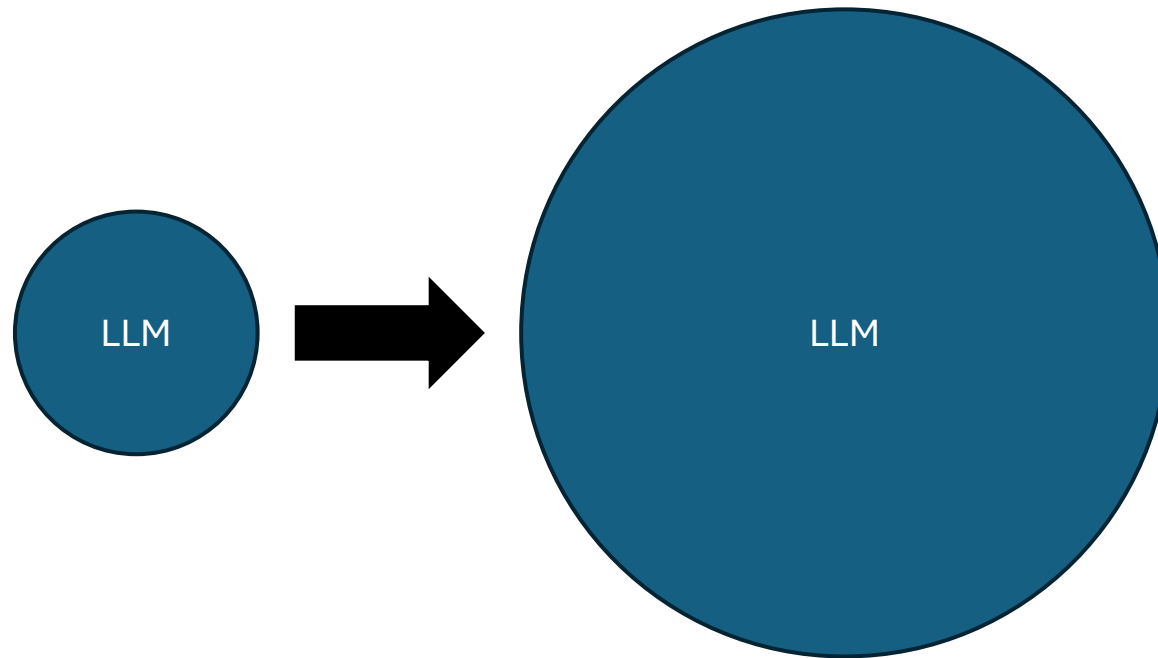
✓ Attention

✓ Parallel training

---

Better RNN for NLP Tasks

# Quality of LLMs: General Reasoning



Detect Hidden Patterns

Common Sense

General Problem Solving

**Reason - הגיון**

# AI agents

An AI agent is a system designed to **perceive its environment**, **reason** and **autonomously act** to achieve specific goals using LLMs.

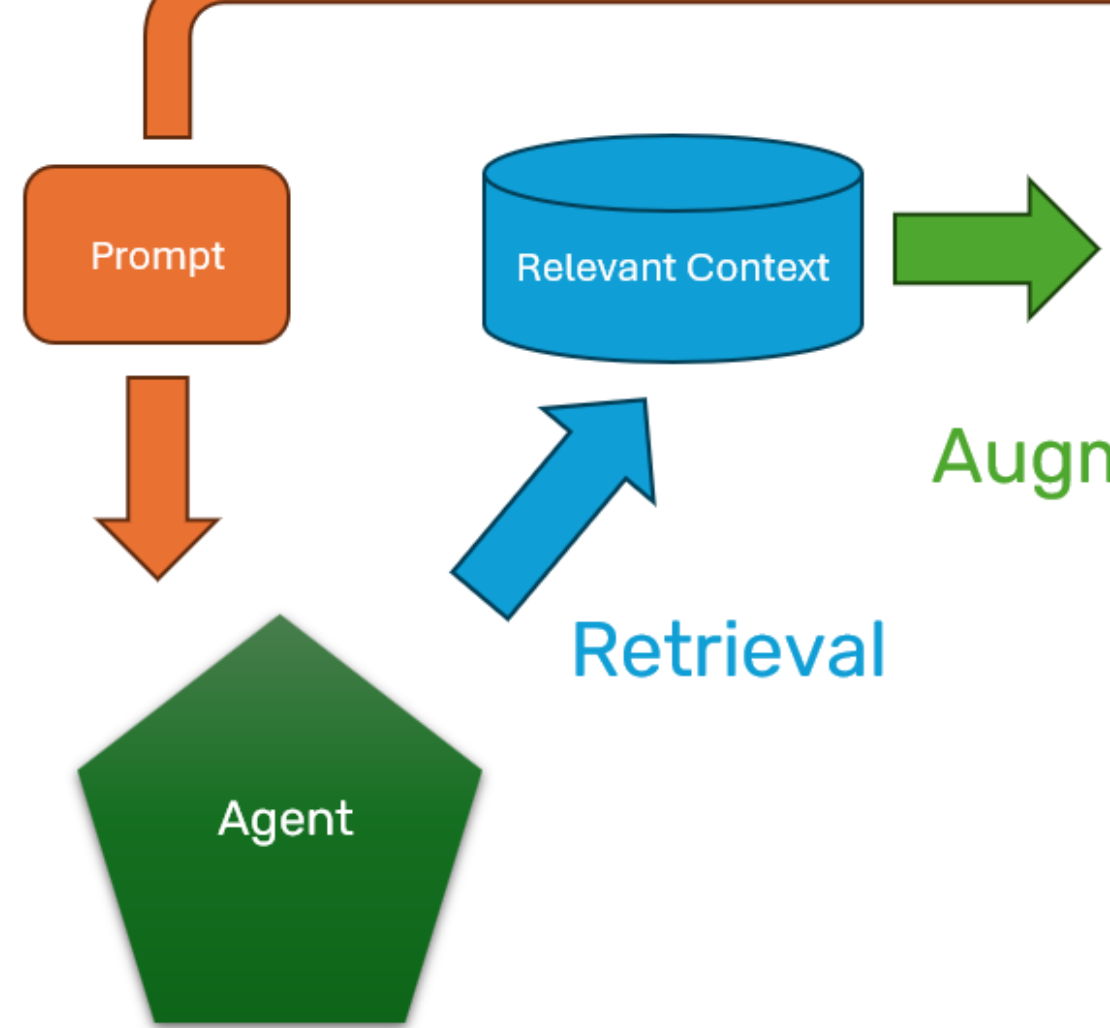


# Agentic RAG

Perceive its environment

Reason

Act Autonomously



# Should I always use Agentic RAG?

## Pros

More relevance/better accuracy

Complex use cases

Scalable

## Cons

Increase latency

Complex

Cost

# Course Submissions

# Course Assignments

| Week | Assignment                                                 | Idea Funnel | Weight                 |
|------|------------------------------------------------------------|-------------|------------------------|
| 3    | Discovering AI Agent Ideas                                 | 3           | Part of course project |
| 4    | Researching Data Needed                                    | 2           | Part of course project |
| 5-6  | Individual assignment                                      |             | 10%                    |
| 7    | AI Agent Technical PRD                                     | 1           | Part of course project |
| 9    | Demo Day Presentation (actual presentations Weeks 10 – 12) |             | Part of course project |

# Course Assignments

## Assignment 1: Discovering AI Agent Ideas

### Goal

Identify **three innovative ideas for AI Agents** that solve **niche, specific problems** in real-world domains.

### Instructions

You will work in **groups of three**, created using the **Moodle group resource** on the course website.

Each group may choose **up to three domains** from this link:

 Domains - Spring 2026

and propose **three AI agent ideas**, each addressing a unique niche problem.

Each domain can be selected by **a maximum of two groups**.

To **reserve** a domain, write your group name next to it in the shared list.

### Submission Format

Each group should submit one short document (PDF or Word) including:

- **Domain Name:**
- **Student Names:**
- **Group Number:**



# Course Assignments

|    | A                               | B                                  | C                                  | D                                  |  |
|----|---------------------------------|------------------------------------|------------------------------------|------------------------------------|--|
| 1  | category                        | domain                             | Team Name 1 (student name A, B, C) | Team Name 2 (student name A, B, C) |  |
| 2  | General Services                | Legal Services                     |                                    |                                    |  |
| 3  | General Services                | Accounting & Finance               |                                    |                                    |  |
| 4  | General Services                | Insurance                          |                                    |                                    |  |
| 5  | General Services                | Consulting                         |                                    |                                    |  |
| 6  | General Services                | Marketing & Advertising            |                                    |                                    |  |
| 7  | General Services                | PR & Communications                |                                    |                                    |  |
| 8  | General Services                | Human Resources (HR)               |                                    |                                    |  |
| 9  | General Services                | Recruiting & Talent Acquisition    |                                    |                                    |  |
| 10 | General Services                | Customer Support / Contact Centers |                                    |                                    |  |
| 11 | General Services                | Business Process Outsourcing (BPO) |                                    |                                    |  |
| 12 | Real Estate & Built Environment | Residential Real Estate            |                                    |                                    |  |
| 13 | Real Estate & Built Environment | Commercial Real Estate             |                                    |                                    |  |
| 14 | Real Estate & Built Environment | Property Management                |                                    |                                    |  |
| 15 | Real Estate & Built Environment | Construction                       |                                    |                                    |  |
| 16 | Real Estate & Built Environment | Architecture & Interior Design     |                                    |                                    |  |
| 17 | Real Estate & Built Environment | Urban Planning                     |                                    |                                    |  |
| 18 | Real Estate & Built Environment | Facility Management                |                                    |                                    |  |
| 19 | Real Estate & Built Environment | Landscaping & Gardening            |                                    |                                    |  |
| 20 | Real Estate & Built Environment | Smart Buildings / PropTech         |                                    |                                    |  |
| 21 | Retail & E-Commerce             | E-Commerce                         |                                    |                                    |  |
| 22 | Retail & E-Commerce             | Retail Stores                      |                                    |                                    |  |
| 23 | Retail & E-Commerce             | Wholesale & Distribution           |                                    |                                    |  |
| 24 | Retail & E-Commerce             | Fashion & Apparel                  |                                    |                                    |  |
| 25 | Retail & E-Commerce             | Consumer Electronics               |                                    |                                    |  |
| 26 | Retail & E-Commerce             | Luxury Goods                       |                                    |                                    |  |
| 27 | Retail & E-Commerce             | Marketplaces                       |                                    |                                    |  |
| 28 | Retail & E-Commerce             | Logistics & Fulfillment            |                                    |                                    |  |

Questions?